ABRA — findings, 2026-07-26

What was measured, what broke, and what it changed. Everything here is reproducible from the repository; every number names the artifact that produced it.

This session's theme, stated plainly: **the self-play corpus was not modelling the format it claims to model, and the reason was a chain of four independent defects, none of which errored.**

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1. The corpus contained no mega evolutions

Champions is a mega-centric format. **93% of real ladder games contain a mega evolution.** Across 199,524 self-play games, ours contained essentially none.

Four separate faults, each sufficient on its own:

1.1 The player was never told it could mega. Showdown's `RandomPlayerAI` does `this.mega = options.mega || 0` — the default is *never* — and `engine/mew.js` never passed the option. Fixed by passing a per-decision probability, which also models the real behaviour: usually the first turn the Pokémon is out, occasionally later. Deliberately not 1.0.

1.2 The teams had no stones to use. `fillSet` consulted Smogon's item list first, and the entry it consulted was the *base forme*, whose item list contains no stones at all (Charizard: Life Orb 41%, Choice Scarf 31%, Charcoal 9%). Our own parse of real Champions replays says Charizardite-Y. An external source was outranking our own direct measurement of this exact format. Our parse now wins; Smogon remains the fallback for species we have not observed enough of.

1.3 Every Tyranitar held a stone. `gearPriors` kept only the *mode*, turning an item from a choice into a species trait. Wrong in both directions — nearly every Charizard really does carry one, plenty of Tyranitars do not. The full distribution is now kept and sampled.

1.4 Megas used the base forme's moves. Mega Dragonite is a special attacker and ordinary Dragonite is physical, and our own priors say exactly that:

	top moves
`dragonite`	Extreme Speed 21%, Earthquake 8% — physical
`dragonitemega`	Roost 25%, Dragon Pulse 12%, Blizzard 12%, Draco Meteor 12% — special

Twenty-six mega formes carry their own priors and every one was ignored, because moves were drawn before the item was known. Unrevealed slots are now re-drawn from the mega forme once a stone is assigned; anything the replay actually revealed is kept.

Result: 54.7% of self-play games now contain a mega, from ~0%.

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2. The bots do not know the type chart

Measured over 20,000 self-play games and 10,740 real ones:

	bots	humans
move hit something **immune**	4.3%	2.2%
move was **super effective**	9.9%	**23.4%**
move **outright failed**	10.3%	2.7%
Protect (all forms)	17.4%	12.1%

Humans land super-effective hits **more than twice as often**. The ratio of super-effective to immune is 10.7 : 1 for humans and 2.3 : 1 for bots — one bot move in twenty-three does literally nothing. The 10.3% "outright failed" is largely repeat Protect.

One cause: the policy samples moves by marginal usage and never looks at the board. It cannot know that Fake Out into a Ghost does nothing, that Ground does nothing to Flying, or that a second Protect will fail.

This is the argument for scoring options against the simulator rather than teaching the bot rules. Score a move by playing it and evaluating the resulting board, and every rule comes free: an immune move changes nothing and scores zero; a failed Protect costs a turn; a super-effective KO scores well. See `docs/THEORY.md` §2 for why the scores must then be played as a **mixed** strategy rather than argmax.

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3. Redundant protection moves

Sets were generated holding both Protect and Detect — 1.1% of Pokémon that used any protection move, across 40,000 games. Co-occurrence lift discourages nonsense pairs but cannot forbid them. Redundant families (protection, weather, terrain) are now capped at one member per set, in both draw paths. Measured after: 0 of 4,800 generated sets.

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4. Smogon's statistics, and whether we may use them the way we assumed

We had been deriving from ~1,700 clean replays things Smogon already publishes from **1,163,315 battles**. Before relying on that, the methodology was checked.

4.1 What the columns mean

The usage files publish four quantities per species: weighted **Usage %**, **Raw** (and its %), and **Real** (and its %). Smogon weights each team by the probability that the player's Glicko rating exceeds the cutoff, so a "1630" file is *weighted toward* strong play — it is **not** a subset of strong players. All four cutoff files for a month report the same total battles, which confirms it.

4.2 The claim we needed to check

We use `raw(Species-Mega) ÷ (raw(Species) + raw(Species-Mega))` as *P(this species carries a mega stone)*. That is only valid if Smogon assigns the mega forme from the **item on the team**, not from whether a mega actually occurred in battle.

It does, and this was verified three independent ways rather than assumed. The question that matters is whether Smogon can know a Pokémon held a stone when it died without ever revealing it. It can, because the statistics are computed **server-side from the team data**, not from replays.

Proof 1 — the raw counts sum to exactly twelve per battle. A VGC battle has 12 Pokémon across the two teams and roughly 8 brought. Over 1,163,315 battles at the 1630 cutoff:

```
sum of RAW counts   13,959,780  ->  12.00 per battle   <- every team slot, brought or not
sum of REAL counts   6,391,567  ->   5.49 per battle   <- only what actually appeared
```

Twelve-point-zero-zero is not an approximation. Every Pokémon on both teams is counted, including ones that never left the back and revealed nothing.

Proof 2 — they publish EV spreads. "Jolly 2/32/0/0/0/32" cannot be inferred from watching a battle; EVs are never revealed. Publishing them at all requires the actual team.

Proof 3 — base-forme Charizard's item list contains zero mega stones (Life Orb 41.4%, Choice Scarf 30.8%, Charcoal 8.7%). If the forme were resolved from battle events, a Charizard that held a stone and died before using it would appear in the base entry holding Charizardite. None do.

The `Raw` / `Real` split is itself the tell: `Raw` is team composition, `Real` is appearances. We want the former, and it is exactly what the mega division uses.

So the split is exactly P(holds a stone), and it is clean for species with two megas:

species	holds a stone	Y	X	no stone
Charizard	**99.3%**	96.1%	3.2%	0.7%
Raichu	88.3%	75.3%	13.0%	11.7%
Swampert	98.0%	—	—	2.0%
Metagross	96.0%	—	—	4.0%
Staraptor	81.7%	—	—	18.3%
Tyranitar	66.0%	—	—	34.0%
Aerodactyl	58.3%	—	—	41.7%
Venusaur	55.5%	—	—	44.5%

Weighted usage, 2026-06, cutoff 1630.

A correction is recorded here. An earlier reading of this session claimed the non-mega share was inflated by Pokémon that carried a stone and died before using it. That is wrong: those Pokémon are counted under the mega forme regardless. The base-forme entries are genuinely stone-less builds.

4.3 Why Smogon beats our own replay parse for this

Our estimate had Tyranitar at 20% and Smogon has it at 66%. Ours is biased, and the mechanism is measurable: an item is only recorded when a replay reveals it, and a mega stone is revealed by the mega itself. In 10,740 real games there are **16,631 `|-mega|` lines against 1,282 `|-item|` lines**. Stones announce themselves; a Tyranitar holding Assault Vest may reveal nothing all game. So our parse systematically overstates stones among *observed* items while understating the stone rate overall, depending on which way the selection cuts.

Smogon sees every team, so there is no reveal bias at all.

4.4 What else is in those files that we were discarding

- **Checks and Counters** — per species, what beats it, with a win rate, a **95% interval**, and how the matchup resolves (KOed% / switched-out%). This is precisely what GURU and COUNTERPLAY tried to derive from ~1,700 replays where nothing reached significance.
- **Teammates** — was truncated to ten, now kept in full. Partner frequency over the whole ladder is a stronger archetype signal than clustering 1,933 observed teams.
- **Viability Ceiling** — the highest GXE achieved by any team using that species.
- **Mega formes as separate species** — the omission behind §1.

4.5 Honest limits on Smogon data

- **Aggregate only.** It describes the population and can never be joined to a specific replay, so it cannot support any per-game claim.
- **Weighting is not filtering.** A 1630 file still contains every player, weighted. Quoting it as "high-ladder only" would be wrong.
- **Teams with high rating uncertainty are excluded** when the baseline is above 1500 (RD > 100), which is a selection we do not control and cannot audit.
- It tells us nothing about **per-turn behaviour**, which is exactly what our replay store is for. The two sources answer different questions and neither replaces the other.

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5. Mega evolution timing

An edge case was raised: can a Pokémon that switches in and is KO'd before acting still mega?

No, and the data confirms the mechanic exactly. Of 16,631 mega events in real games, **zero** occurred on the Pokémon's switch-in turn — switching *is* that turn's action. Separately, 1.88% of switch-ins faint on the turn they arrive and never act at all.

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6. Seed space

At millions of games the seed scheme needed checking. Three things were measured:

1. **Real and fixed.** The farm's seed base was `Date.now() % 1e7`, which **wraps every 2.78 hours**. Two runs started that far apart drew the same base and produced overlapping ranges — the same battles regenerated under fresh ids, the one corruption a duplicate-id check cannot see.
2. **Real, documented, guarded.** The engine uses only **28 bits** of the seed: `[s&0xffff, (s>>4)&0xffff, (s>>8)&0xffff, (s>>12)&0xffff]` discards everything above bit 27, so any two seeds 268,435,456 apart are the same battle. Far beyond Metamon scale (20M), so a ceiling rather than a live problem — but the farm now refuses to cross it.
3. **Not real.** Adjacent seeds do **not** produce similar battles: seeds N and N+1 on identical teams diverge at the first protocol line.

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7. Corpus and site integrity

- **Side bias, fixed.** The matchup enumeration emitted pairs as (low index, high index) and always sent the low index to p1, so p1 won 50.86% of 119,826 non-mirror games, 95% CI [50.58, 51.15] — excluding 50. Mirrors showed nothing, confirming the harness was fair and the pairing was not. After the fix: 50.12%, interval covering 50.
- **The validator could not read a large corpus.** `readFileSync` on a 988MB store exceeds V8's ~512MB string cap, so the tool that exists to certify big corpora was guaranteed to fail on one.
- **Dataset counts are generated.** Six different hardcoded sizes were live across six rooms. All now derive from `data/live.js`, written by `engine/refresh-site-data.py` from the same filter every engine uses (S13).
- **Turns per game was measured on the wrong population** — all 12,872 collected games, three quarters of them bot games, in a tile beside one computed on clean games only. Corrected to 8.3 turns over clean games, from 5.4.

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8. What this means for the corpus

The 199,524-game corpus is superseded. Every game in it was played with no megas, wrong items, and mega Pokémon using base-forme moves. It remains mechanically valid — the engine was always correct — but the *teams* were not the teams of this format, so nothing metagame-flavoured should be drawn from it. Regenerate before further analysis.

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9. Open

- The **transfer test** (train on self-play, evaluate on real games) has not completed.
- The **scoring bot** of §2 is designed but not built.
- GURU still uses its own underpowered matrix rather than Smogon's checks-and-counters.
- Mega rates from §4.2 are parsed but not yet driving set construction.

Sources

- Smogon usage statistics methodology — Weighted Stats FAQ, Usage Statistics Discussion, Viability Ceiling
- Archived statistics: `data/smogon-stats/2026-06/{usage,moveset}/`
- Parser and priors: `engine/smogon_priors.js`, `engine/set_priors.js`
- Corpus tooling: `engine/mew.js`, `engine/mew_farm.js`, `engine/validate_selfplay.js`